Lecture 9 Practice Questions and Solutions

Question 1 (q1_sol.py)

Write a function called print_in_box(word) that prints word in a box as shown. Use the print_in_bars function from the previous question to print the middle line of the box. Then ask the user to enter a movie name and print in a box.

```
Enter a movie name: The Matrix
+----------+
|The Matrix|
+----------+
```

Sample Solution

```
def print_in_bars(word):
    print(f'|{word}|')

def print_in_box(word):
    bar = '+' + '-' * len(word) + '+'
    print(bar)
    print_in_bars(word)
    print(bar)


movie = input("Enter a movie name: ")
print_in_box(movie)
```

Question 2 (q2_sol.py)

Write a program that asks the user to enter an int that is 0 or greater, and a word, and then prints the word in a numbered list as shown. Use a for-loop in your solution. No functions are required, but you can make one if you want.

```
Enter an int: 3
Enter a word: hello!
hello!
hello!
hello!
```

Sample Solution

```
n = int(input("Enter an int: "))
word = input("Enter a word: ")
print()
for i in range(n):
    print(f'{i+1}. {word}')
```

Question 3 (q3_sol.py)

Write a program that asks the user to enter an int (called n)that is 0 or greater. Then, starting at 1, print the ints from 1 to n on the screen in a box as shown. Use the print_in_box(word) function from the previous question.

```
Enter an int: 3

+-+
|1|
+-+

+-+
|2|
+-+

+-+
|3|
+-+
```

Sample Solution

```
def print_in_bars(word):
    print(f'|{word}|')

def print_in_box(word):
    bar = '+' + '-' * len(word) + '+'
    print(bar)
    print_in_bars(word)
    print(bar)

n = int(input("Enter an int: "))
print()
for i in range(1, n+1):
    print_in_box(str(i))
    print()
```

Question 4 (q4_sol.py)

Write a program that asks the user to enter an int that is greater than 1, and then prints a countdown from that int to 1, finally printing "Blastoff!".

```
Enter an int: 5
5
4
3
2
1
Blastoff!
```

Sample Solution

```
start = int(input("Enter an int: "))
for i in range(start):
    print(start - i)
print("Blastoff!")
```